Ali Nazari

(+49) 1525/3531738 — ali.nazari.8102@gmail.com — linkedin.com/in/ali-nazari — github.com/nazari02105 — Google Scholar

Summary — I am a first-year M.Sc. student in Computer Science at Dresden University of Technology, with a B.Sc. in Computer Engineering from Sharif University of Technology and research experience in Machine Learning. I have co-authored five research papers and collaborated with international research groups in Germany, Denmark, and the United States. My research interests include Natural Language Processing, Vision-Language Models, and Trustworthy Machine Learning, with a focus on developing and evaluating reliable, efficient AI systems.

Education

Dresden University of Technology — **October 2026 – Present**
Master of Science - Computer Science — Dresden, Germany
Overall GPA: Not yet available (first semester)

Sharif University of Technology — **September 2020 – July 2025**
Bachelor of Science - Computer Engineering — Tehran, Iran
Overall GPA: 19.16/20.00 Major GPA: 19.42/20.00

Abu Ali Sina High School — **June 2017 – July 2020**
Diploma - Mathematics and Physics — Tehran, Iran
Overall GPA: 19.86/20.00 National university entrance exam rank: 60/155000

Research Interests

Natural Language Processing **Vision-Language Models** **Trustworthy ML** **Representation Learning**

Publications

- Shahariar, GM., Shayegani, E., **Nazari, A.**, Abu-Ghazaleh, N. "Modeling Hierarchical Thinking in Large Reasoning Models." *Accepted as a Spotlight paper at the **ICML 2026 Conference**, May 2026.*
- Asgari, E., El Kheir, Y., Sadraei Javaheri, M.A., **Nazari, A.** "MorphBPE: Morphology-Aware Tokenization for Efficient LLM Training." *Accepted to **Findings of ACL 2026 Conference**, April 2026.*
- **Nazari, A.**, Najafi, S.M.Y., Abbasi, R., Mohammad-Rahimi, H., Motie, P., Iranparvar Alamdari, M., Hosseinzadeh, M., Pauwels, R., Schwendicke, F. "Deep Learning-based Super-Resolution for Dentomaxillofacial Cone-Beam Computed Tomography: A Pilot Study." *Accepted to the **Imaging Science in Dentistry Journal**, July 2025.*
- Abbasi, R., **Nazari, A.**, Sefid, A.R., Banayeeanzade, M.A., Rohban, M.H., Soleymani-Baghshah, M. "CLIP Under the Microscope: A Fine-Grained Analysis of Multi-Object Representation." *Accepted to the **CVPR 2025 Conference**, March 2025.*
- Abbasi, R., **Nazari, A.**, Sefid, A.R., Banayeeanzade, M.A., Rohban, M.H., Soleymani-Baghshah, M. "Analyzing CLIP's Performance Limitations in Multi-Object Scenarios: A Controlled High-Resolution Study." *Accepted to the **ECCV 2024 Workshop EVAL-FoMo**, August 2024.*

Research Experience

Safety and Reasoning Analysis of Large Reasoning Models — **May 2025 - May 2026**
Supervisor: Prof. Nael Abu-Ghazaleh — University of California, Riverside, United States

Investigated chain-of-thought trajectories in Large Reasoning Models using a six-state Finite State Machine (FSM) to identify transitions associated with correct and incorrect solutions. Focused on FSM trajectory analysis and developing a training-free, Q-value-guided activation-steering method to improve reasoning across mathematical and scientific benchmarks with fewer interventions.

Analyzing CLIP's Performance Limitations in Multi-Object Scenarios — **February 2024 - April 2025**
Supervisors: Prof. Mohammad Hossein Rohban and Prof. Mahdieh Soleymani — Sharif University of Technology, Iran

Investigated CLIP's biases in multi-object scenarios, including its preference for earlier-mentioned objects in text and larger objects in images, and their effects on retrieval, classification, image-text matching, and text-to-image generation. Contributed to constructing the ComCO dataset and was primarily responsible for all text-encoder experiments, analyzing positional sensitivity and attention behavior and evaluating the impact of these biases on downstream tasks.

Image Retrieval for Medical Imaging **November 2023 - February 2024**
Supervisor: Prof. Mohammad Hossein Rohban Sharif University of Technology, Iran

Improved medical image retrieval performance through a semi-supervised approach and developed a novel re-ranking method combining bi-encoder and cross-encoder architectures. Identified a critical validation bug during comparative evaluation of DINO and a supervised baseline, helping ensure reliable assessment of model performance.

Protein Large Language Model **July 2023 - November 2023**
Supervisor: Prof. Alice Carolyn McHardy Helmholtz Center for Infection Research, Germany

Developed a protein language model for antibiotic-resistance prediction across bacterial species, implementing custom BPE tokenization and efficient distributed training with FlashAttention and DeepSpeed, while exploring morphology-aware tokenization strategies to improve LLM training across diverse languages.

Enhancing 3D Medical Imaging Resolution and Quality Using Deep Learning Models **March 2023 - July 2023**
Supervisor: Dr. Hossein Mohammad-Rahimi Aarhus University, Denmark

Developed deep learning-based super-resolution models for dentomaxillofacial CBCT images using a 3D dental dataset, simulating image degradation through downscaling, blurring, and noise injection.

Teaching Experience

In all the courses listed below, I was part of the team that designed and graded projects and assignments and conducted workshops.

- ***Natural Language Processing*** *- Dr. Ehsaneddin Asgari (Spring 2025)*
- ***Machine Learning*** *- Dr. Seyed Abolfazl Motahari (Spring 2024 & Fall 2023) - Dr. Mahdi Jafari Siavoshani (Fall 2023)*
- ***Artificial Intelligence*** *- Dr. Mohammad Hossein Rohban (Spring 2024 & Spring 2023)*
- ***Modern Information Retrieval*** *- Dr. Mahdieh Soleymani (Spring 2024) - Dr. Hamid Beigy (Fall 2023)*

Industry Experience

AI Engineer ***May 2026 - October 2026***
Built an agentic AI contract assistant (Icertis/Harvey-style) and an AI-powered BI dashboard. *Avitec, Iran*
Skills: *Python, LangGraph, LangChain, Agentic AI, LLMs, RAG, Document Summarization, Legal Contract Analysis*

AI Engineer ***May 2025 - May 2026***
Developed an agentic AI platform for financial services with voice-enabled interaction. *Hesabo, Iran*
Skills: *Python, LangGraph, LangChain, Agentic AI, LLMs, Django, Speech-to-Text, Text-to-Speech*

Data Engineering Internship ***July 2022 - December 2022***
The main focus was on software and data engineering using the C# language and the ASP.NET framework. *Code Star, Iran*
Skills: *C#, ASP.Net, ETL, ORM, Ontology, Node Expand, Elastic Search, Spark*

Extracurricular Activities

Winter Seminar Series (WSS) ***2024***
Talks on cutting-edge CS topics by global researchers and professors - Technical Staff *Sharif University of Technology, Iran*

Hardware Challenge ***2024***
A competition on hardware, FPGA, and IoT - Executive Staff *Sharif University of Technology, Iran*

Skills

Programming & Data	*Python, C++, Java, SQL, NumPy, pandas*
Deep Learning & Fine-Tuning	*PyTorch, TensorFlow, Hugging Face Transformers, PEFT (LoRA/QLoRA), bitsandbytes, Unsloth*
Training & Inference	*DeepSpeed, Hugging Face Accelerate, FlashAttention, Ray, vLLM, SGLang, Ollama, TensorRT-LLM*
RAG, Agents & Retrieval	*LangChain, LangGraph, LlamaIndex, Haystack, Sentence Transformers, Chroma, Qdrant, Pinecone, Elasticsearch*
Evaluation & Tracking	*lm-evaluation-harness, Hugging Face Evaluate, DeepEval, Weights & Biases, MLflow, TensorBoard*
Development Tools	*Linux, Git, Docker, LaTeX*

Languages

Persian: *Native* ***English:*** *C1 (TOEFL iBT: 108/120)* ***German:*** *A1*